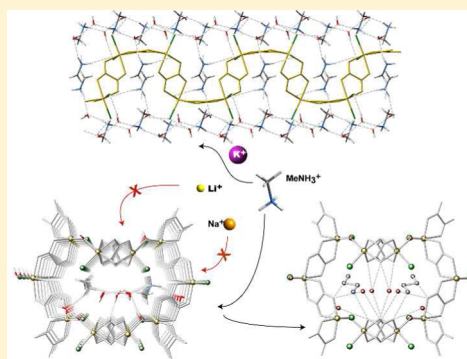


Selective Guest Inclusion in Oxalate-Based Iron(III) Magnetic Coordination Polymers

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S Supporting Information

ABSTRACT: The preparation and structural characterization of four novel oxalate-based iron(III) compounds of formulas $\{(\text{MeNH}_3)_2[\text{Fe}_2(\text{ox})_2\text{Cl}_4] \cdot 2.5\text{H}_2\text{O}\}_n$ (1), $\text{K}(\text{MeNH}_3)[\text{Fe}(\text{ox})\text{Cl}_3(\text{H}_2\text{O})]$ (2), $\{\text{MeNH}_3[\text{Fe}_2(\text{OH})(\text{ox})_2\text{Cl}_2] \cdot 2\text{H}_2\text{O}\}_n$ (3), and $\{(\text{H}_3\text{O})(\text{MeNH}_3)[\text{Fe}_2\text{O}(\text{ox})_2\text{Cl}_2] \cdot 3\text{H}_2\text{O}\}_n$ (4) (MeNH_3^+ = methylammonium cation and H_2ox = oxalic acid) are reported here. 1 is an anionic waving chain of oxalato-bridged iron(III) ions with peripheral chloro ligands, the charge balance being ensured by methylammonium cations. 2 is a mononuclear complex with a bidentate oxalate, three terminal chloro ligands, and a coordinated water molecule achieving the six-coordination around each iron(III) ion. Its negative charge is balanced by potassium(I) and methylammonium cations. 3 and 4 are made up of oxalate-bridged and either hydroxo (3)- or oxo-bridged (4) iron(III) chiral three-dimensional (3D) networks of formulas $[\text{Fe}_2(\text{OH})(\text{ox})_2\text{Cl}_2]_n^{2-}$ (3) and $[\text{Fe}_2\text{O}(\text{ox})_2\text{Cl}_2]_n^{2-}$ (4) with methylammonium (3 and 4) and hydronium (4) as counterions. The common point these compounds share is related to their synthetic strategy, which consists of the use of mixed alkaline/alkylammonium cations as templating agents for the growth of the 1D or 3D iron(III) motifs. Interestingly, even in the presence of any given alkaline cation in the reaction solutions, the resulting coordination polymers (1, 3, and 4) exclusively contain the methylammonium cation, revealing the highly selective character of the 1D and 3D networks. Furthermore, the isolation of the very unstable compound 1 could be only achieved in the presence of the KCl salt, suggesting a probable templating effect of the potassium(I) cations. Finally, a study of the variable-temperature magnetic properties of the 3D compounds 3 and 4 showed the occurrence of weak ferromagnetic ordering due to a spin canting, the value of the critical temperature (T_c) being as high as 70 K.



■ INTRODUCTION

Porous coordination polymers (PCPs) are excellent candidates for the design and synthesis of new functional materials.^{1–5} In particular, their fascinating host–guest chemistry^{6–9} offers endless possibilities concerning the design of multifunctional materials.^{10–19} Because of their porous character, PCPs have proved to be useful in catalysis,²⁰ chemical sensing,^{21,22} and gas storage and separations.^{23–25} The desired network topology, which undoubtedly influences the final physical properties, can be predesigned by choosing appropriate synthetic routes and suitable metal ions and bridging or ancillary ligands. In this respect, simply focusing on the vast possibilities offered by the evergreen oxalate ligand, a seemingly infinite number of multifunctional materials have been synthesized.^{26,27} In these compounds, the anionic networks of general formula $[\text{M}^{\text{I/II}}\text{M}^{\text{III}}(\text{ox})_3]^{2-/-}$ ($\text{M} = \text{Cr}, \text{Fe}$ and $\text{ox} = \text{oxalate}$) act as hosts of a great variety of either organic or inorganic cations as well as cationic metal complexes to afford hybrid compounds.^{28–30} The long-range magnetic ordering occurring in some of them, which is provided by the oxalate framework,³¹ can coexist with other properties such as chirality,^{32–37}

magneto-chiral dichroism,³⁸ spin transition,^{39–43} proton conduction,^{44–47} electric conductivity,^{48–50} ferroelectricity,⁵¹ and second-order optical nonlinearity.^{52–55}

Thinking of the role that can be played at the synthetic level by subtle factors such as the type of counterion, oxalate to metal ion molar ratio, and crystallization time lapse, some of us have recently reported a strategy for the selective self-assembly of oxalate-bridged 1D and oxalato-/oxo-bridged 3D iron(III) systems.^{56–60} As a result, we identified the dimethylammonium cation (Me_2NH_2^+) as an organic cation which is able to discriminate between the formation of the oxalate-bridged 1D and the oxalato-/oxo-bridged 3D polymers as well as established the appropriate synthetic conditions for the selective preparation of the antiferromagnetically coupled 1D motifs or the spin-canted 3D chiral networks.^{57,59} Among the 3D compounds, two slightly different oxalato-/oxo-bridged iron(III) compounds of formulas $\{(\text{EtNH}_3)_2[\text{Fe}_2\text{O}(\text{ox})_2\text{Cl}_2] \cdot 2\text{H}_2\text{O}\}_n$ and $\{(\text{H}_3\text{O})(\text{EtNH}_3)[\text{Fe}_2\text{O}(\text{ox})_2\text{Cl}_2] \cdot \text{H}_2\text{O}\}_n$ (EtNH_3^+

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Table 1. Crystal Data and Structure Refinement Details for Compounds 1–4

	1	2	3	4
empirical formula	C ₁₂ H ₃₄ Cl ₈ Fe ₄ N ₄ O ₂₁	C ₃ H ₈ Cl ₃ FeKNO ₅	C ₅ H ₁₃ Cl ₂ Fe ₂ NO ₁₂	C ₅ H ₁₃ Cl ₂ Fe ₂ NO ₁₂
Z	2	2	4	8
T, K	100	293	100	100
fw	1077.43	339.40	461.76	461.76
cryst syst	monoclinic	monoclinic	monoclinic	orthorhombic
space group	P2 ₁ /c	P2 ₁ /m	Cc	Fdd2
a, Å	10.7837(7)	8.725(2)	8.945(3)(4)	14.914(4)
b, Å	13.9135(10)	7.508(2)	14.936(5)	23.507(8)
c, Å	13.0578(9)	9.026(3)	12.408(5)	9.261(4)
β, deg	104.060(4)	112.421(7)	108.531(8)	
V, Å ³	1900.4(2)	546.5(3)	1571(7)	3247(2)
D _c , g cm ^{−3}	1.883	2.063	1.951	1.889
μ, mm ^{−1}	2.137	2.487	2.242	2.171
R1 (I > 2σ(I)) ^a	0.0337	0.0391	0.0556	0.0382
wR2 ^{b,c}	0.0769	0.1021	0.1557	0.1048

^aR1 = $\sum ||F_o| - |F_c|| / \sum |F_o|$. ^bwR2 = $\{\sum [w(F_o^2 - F_c^2)^2] / [\sum (w(F_o^2)^2)]\}^{1/2}$. ^cw = $1 / [\sigma^2(F_o^2) + (aP)^2 + bP]$ with $P = [F_o^2 + 2F_c^2] / 3$; a = 0.0283 (1), 0.0593 (2), 0.0927 (3), 0.0794 (4); b = 0.8357 (1), 0.1726 (2), 10.7937 (3), 4.8024 (4).

= ethylammonium cation) were achieved by using the same cation but changing the iron(III) to counterion molar ratio.^{58,59}

The lowering of the symmetry of the cavities induced by the presence of two different counterions (H₃O⁺ and EtNH₃⁺) in the last compound versus only one type (EtNH₃⁺) in the first compound seems to be at the origin of the increase of the T_c value (from 56 to 70 K). This hypothesis suggested a new strategy to prepare oxalato-/oxo-bridged 3D iron(III) compounds with high T_c values by the use of mixed cations to balance the negative charge of the 3D network.

Hence, we planned the use of mixed alkaline/alkylammonium cations as templating agents in the synthetic route of the oxalato-/oxo-bridged 3D iron(III) networks. In so doing, FeCl₃ was reacted with oxalic acid (H₂ox) in the presence of the methylammonium ion (MeNH₃⁺) and K⁺, Na⁺, or Li⁺ as chloride salts. The new and very unstable (vide infra) chain $\{(\text{MeNH}_3)_2[\text{Fe}_2(\text{ox})_2\text{Cl}_4] \cdot 2.25\text{H}_2\text{O}\}_n$ (1) and the mononuclear species $\text{K}(\text{MeNH}_3)[\text{FeCl}_3(\text{ox})(\text{H}_2\text{O})]$ (2) were isolated together from all reaction mixtures in the presence of the KCl salt. In turn, the 3D compound of formula $\{\text{MeNH}_3[\text{Fe}_2(\text{OH})(\text{ox})_2\text{Cl}_2] \cdot 3\text{H}_2\text{O}\}_n$ (3), was persistently obtained when NaCl or LiCl was used. Remarkably, in addition 3 is not stable for more than 2 days when it is separated from the mother liquor and undergoes a crystal to crystal transformation evolving to the final product of the reaction, which is a compound of formula $\{(\text{H}_3\text{O})(\text{MeNH}_3)[\text{Fe}_2\text{O}(\text{ox})_2\text{Cl}_2] \cdot 2\text{H}_2\text{O}\}_n$ (4). A similar irreversible crystal to crystal transformation has been already observed for the ethylammonium analogue,⁵⁵ where an intermolecular proton transfer reaction occurring within the net was detected. However, the process in this latter case took place more slowly.

The present work concerns the preparation and structural determination of complexes 1–4 together with a magnetic study of the spin-canted systems 3 and 4 in order to evaluate their T_c values. It is worth noting that, even in the presence of any of the alkaline cations in the reaction solutions, all of the polymeric compounds exclusively contain the methylammonium cation, revealing the highly selective character of both 1D and 3D networks. Furthermore, the isolation of the very unstable compound 1 was only achieved in the presence of the KCl salt, suggesting that there is a probable templating effect of K⁺ ions and, more widely, that the presence of electrolytes

influences the involved chemical equilibria, favoring the formation of minor products.

EXPERIMENTAL SECTION

Materials. All chemicals were purchased from commercial sources and used as received without further purification. Elemental analyses (C, H, and N) were performed by the Microanalytical Service of the Università della Calabria.

Preparation of the Complexes. $\{(\text{MeNH}_3)_2[\text{Fe}_2(\text{ox})_2\text{Cl}_4] \cdot 2.5\text{H}_2\text{O}\}_n$ (1) and $[\text{K}(\text{MeNH}_3)[\text{FeCl}_3(\text{ox})(\text{H}_2\text{O})]]$ (2). A mixture of X-ray-quality green plates (1) and yellow parallelepipeds (2) was separated from an aqueous solution (10 mL) containing FeCl₃ (5 mmol), H₂ox (5 mmol), MeNH₃Cl (3 mmol), and KCl (20 mmol) by slow evaporation at room temperature within 2 weeks. They were filtered off, dried on filter paper, and separated by hand. Yield: 60% (1) and 30% (2). Anal. Calcd for C₁₂H₃₄Cl₈Fe₄N₄O₂₁ (1): C, 13.38; H, 3.18; N, 5.20. Found: C, 13.60; H, 3.32; N, 5.10. Calcd for C₃H₈Cl₃FeKNO₅ (2): C, 10.62; H, 2.38; N, 4.13. Found: C, 10.50; H, 2.00; N, 4.10. Crystals of 1 are hygroscopic and dissolve after 2 min at room temperature on separation from the mother liquor, affording a yellow polycrystalline powder of 3 in a few days.

$\{\text{MeNH}_3[\text{Fe}_2(\text{OH})(\text{ox})_2\text{Cl}_2] \cdot 3\text{H}_2\text{O}\}_n$ (3). Compound 3 could be obtained as a unique product by slow evaporation at room temperature of an aqueous solution (10 mL) containing FeCl₃ (5 mmol), H₂ox (5 mmol), MeNH₃Cl (3 mmol), and XCl (20 mmol, X = Li⁺, Na⁺). Yellow rhombuses of 3 suitable for X-ray diffraction were grown within 1 week. The formation of 1 was not observed under these conditions (without the addition of KCl salt to the solution). Yield: ca. 65% (1). Anal. Calcd for C₅H₁₃Cl₂Fe₂NO₁₂ (3): C, 13.01; H, 2.84; N, 3.03. Found: C, 13.13; H, 2.96; N, 3.10. Alternatively, after complete dissolution of crystals of 1 in a beaker, because of their hygroscopic character, single crystals of 3 were formed in the subsequent drying process at room temperature within 2 weeks.

$\{(\text{H}_3\text{O})(\text{MeNH}_3)[\text{Fe}_2\text{O}(\text{ox})_2\text{Cl}_2] \cdot 2\text{H}_2\text{O}\}_n$ (4). X-ray-quality crystals of 4 have been obtained from single crystals of 3 that undergo an irreversible crystal to crystal transformation which is practically complete at room temperature after 3 days without any loss of crystallinity (see Figure S1 in the Supporting Information). Yield: ca. 65%. Anal. Calcd for C₅H₁₃Cl₂Fe₂NO₁₂ (4): C, 13.01; H, 2.84; N, 3.03. Found: C, 12.95; H, 2.92; N, 2.89.

Physical Techniques. Variable-temperature (1.9–295 K) magnetic susceptibility and magnetization measurements on polycrystalline samples of 3 and 4 (amounting to ca. 11 mg) were carried out with a Quantum Design SQUID susceptometer in the presence of applied magnetic fields ranging from 100 G to 5 T. The alternating current (ac) measurements on polycrystalline samples of 3 and 4 were carried

out at frequencies ranging from 33 to 1000 Hz with an ac field amplitude of 1 G and in the absence of an external magnetic field. The experimental susceptibility data were corrected for the diamagnetic contribution of the constituent atoms and also for the sample holder.

X-ray Crystallographic Analysis. X-ray diffraction data of 1–4 were collected with a Bruker Nonius APEXII CCD area detector diffractometer. Graphite-monochromated Mo $K\alpha$ radiation ($\lambda_{K\alpha} = 0.71073$ Å) was used in all cases. The data collections on single crystals of 1, 3, and 4 were performed at low temperature ($T = 100$ K). In particular, a single crystal of 1 was selected and mounted on a MITIGEN holder, sealed in Paratone oil, and very quickly placed under a dried liquid nitrogen stream cooled to 100 K to avoid rapid degradation, and then the data collection was carried out in a fast manner with an exposure time of 1 s/frame and was complete in less than 2 h. The data collection on compound 4 started when transformation of the single crystal of 3 was practically complete. The data were processed through the SAINT⁶¹ reduction and SADABS⁶² absorption software. The structures were solved by direct methods and subsequently completed by Fourier recycling using the SHELXTL software package.^{63,64} A good model for the cations and solvent molecules was found in the refinement process for 1 and 3. All the non-hydrogen atoms in both compounds were refined anisotropically, with the exception of the O(3w) atom in 1. The occupancy factor of this atom was set to a value of 0.50, in agreement with the corresponding electron density found on a ΔF map.

The hydrogen atoms of this water molecule were not defined, while all the hydrogen atoms on the other water molecules were located on a ΔF map and refined with restraints. The hydrogen atoms of the MeNH_3^+ cations in 1 and 3 and of the hydroxo bridge in 3 were set in calculated positions and refined as riding atoms; statistical disorder accounted for Alert A in checkcif. The cations and the water molecules in compound 4 show large thermal motions because of the disordered rearrangement that follows the single-crystal to single crystal transformation. Consequently, these atoms were not refined anisotropically and their hydrogen atoms were not defined. Different residual peaks of electron density were found in the refinement of the structure of 3. They are located near the iron(III) centers and the chlorine atoms and they are most likely due to the occurrence of absorption. Full-matrix least-squares refinements on F^2 for all compounds were carried out by minimizing the function $\sum w(|F_o| - |F_c|)^2$, and they reached convergence with values of the discrepancy indices shown in Table 1. The graphical manipulations were performed with the CRYSTAL MAKER⁶⁵ software. A summary of the crystallographic data for 1–4 is given in Table 1, whereas selected bond lengths and angles are given in Tables S1–S4 in the Supporting Information for 1–4, respectively.

RESULTS AND DISCUSSION

Description of the Structures. $\{(\text{MeNH}_3)_2[\text{Fe}_2(\text{ox})_2\text{Cl}_4] \cdot 2.25\text{H}_2\text{O}\}_n$ (1). Compound 1 is isostructural with the already known chain of formula $\{(\text{Me}_2\text{NH}_2)_2[\text{Fe}_2(\text{ox})_2\text{Cl}_4] \cdot \text{H}_2\text{O}\}_n$.⁵⁹ Waving $[\text{Fe}_2(\text{ox})_2\text{Cl}_4]^{2n-}$ anionic chains, MeNH_3^+ cations, and crystallization water molecules are present in the structure of 1 (Figure 1). The iron(III) ions of these compounds are bridged by bis-bidentate oxalate groups, with two chlorine atoms acting as terminal ligands. Electrostatic interactions as well as hydrogen bonds involving the counterions, the oxalate groups, the coordinated chlorine atoms, and the crystallization water molecules ensure the cohesion of the crystal lattice (Figure S2 in the Supporting Information). Two crystallographically independent iron(III) ions (Fe(1) and Fe(2)) occur in the asymmetric unit of 1. They are both six-coordinate in distorted-octahedral environments, each one being bonded to four oxygen atoms of two *cis* oxalate groups and to two chlorine atoms. The Fe–O(ox) (values varying in the range 2.025(2)–2.145(2) Å) and Fe–Cl bond lengths (2.2531(6)–2.2795(6) Å) are in agreement with those reported for the other similar

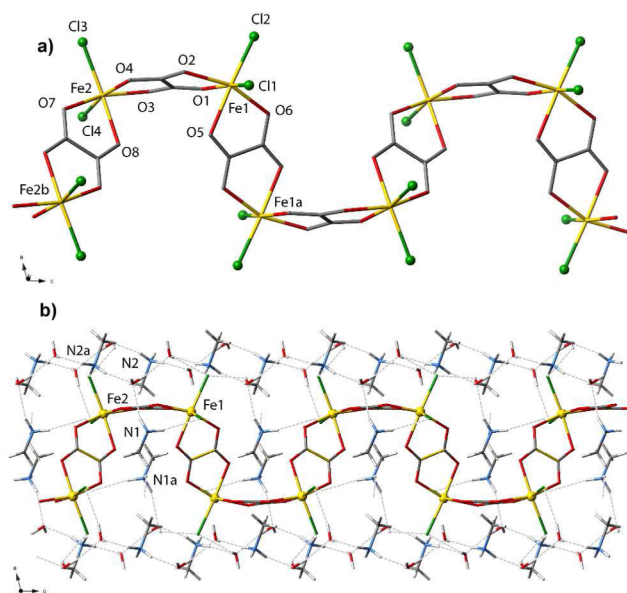


Figure 1. (a) Crystal structure of a fragment of the $[\text{Fe}_2(\text{ox})_2\text{Cl}_4]^{2n-}$ chain in 1 growing along the crystallographic c axis. (b) Details of the hydrogen bonds involving the MeNH_3^+ ions, the oxalate groups, the coordinated chloro atoms, and the crystallization water molecules. Color code: iron (yellow), chlorine (green), oxygen (red), nitrogen (pale blue), carbon (gray), and hydrogen (pale gray).

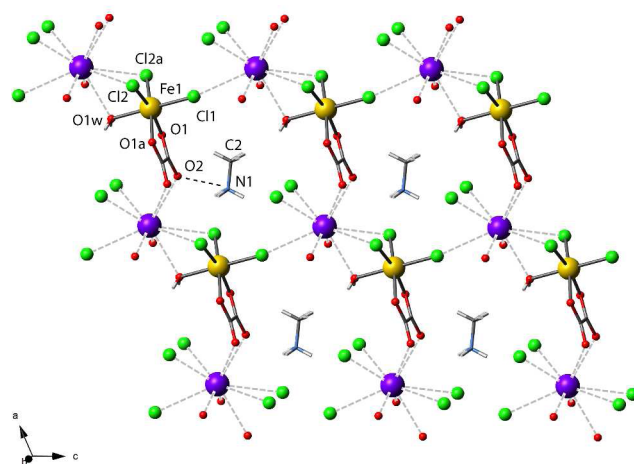


Figure 2. View of a fragment of the structure of 2 showing the interactions between the complex anions and the potassium(I) cations. Color code: potassium (purple), iron (yellow), chlorine (green), oxygen (red), nitrogen (pale blue), and carbon (gray). Symmetry code: (a) $x, -y + 3/2, z$.

oxalato-bridged iron(III) polymers.^{56–60} The main distortion from the ideal octahedral geometry of the metal environment is due to the reduced bite angle of the oxalate ligand (values varying in the range 78.72(6)–79.17(6)°). The iron–iron separations through the three crystallographically independent oxalate ligands are 5.394(2) Å (Fe(1)⋯Fe(2)), 5.458(2) Å (Fe(1)⋯Fe(1a); symmetry code (a) = $1 - x, -y, 1 - z$), and 5.478(2) Å (Fe(2)⋯Fe(2b); (b) = $1 - x, -y, 2 - z$), values that compare well with those observed in the previously reported parent chainlike compounds.

The main structural feature of 1 concerns the arrangement of the chains in the packing, which essentially differs from the usual zigzag motif of the similar 1D compounds. The oxalate

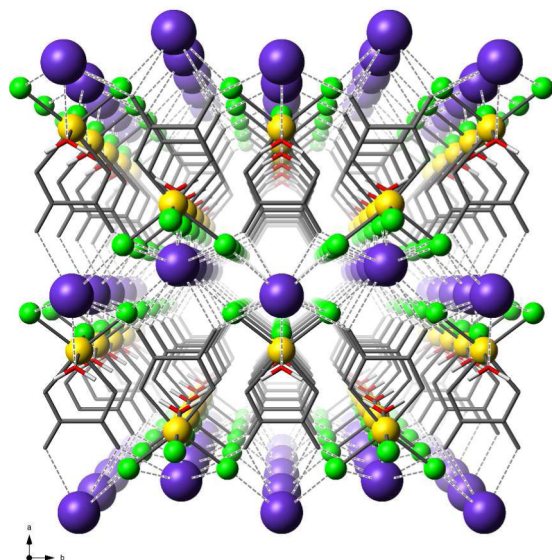


Figure 3. View along the crystallographic *c* axis of a fragment of the crystal packing in **2**. Color code: potassium (purple), iron (yellow), chlorine (green), oxygen (red), and carbon (gray).

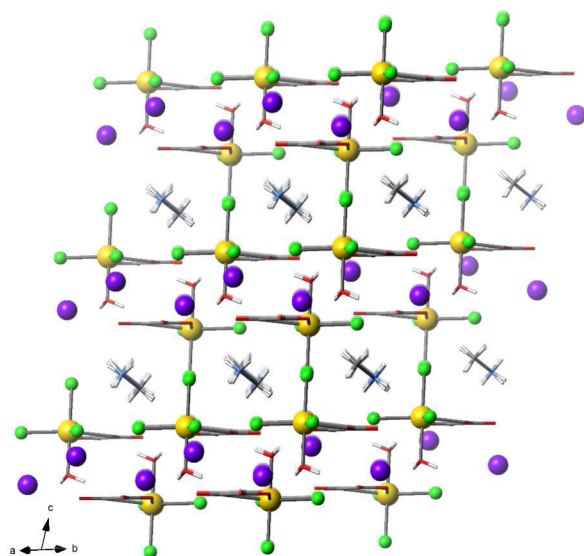


Figure 4. View of the 3D supramolecular motif in **2** which is built by the mononuclear iron(III) units and the potassium(I) ions, showing the channels where the methylammonium cations are hosted. Color code: potassium (purple), iron (yellow), chlorine (green), oxygen (red), nitrogen (pale blue), and carbon (gray).

groups in **1** produce a waving chain motif developing along the crystallographic *c* axis. Focusing on the oxalate that bridges the Fe(1) and Fe(2) atoms, the Cl(1) and Cl(4) atoms are in *trans* positions whereas Cl(2) and Cl(3) are placed in *cis* positions. However, the chloride ligands on adjacent iron(III) ions are in *trans* positions with respect to the other two crystallographically independent oxalate bridges.

To the best of our knowledge, a similar motif has been observed only in the parent dimethylammonium derivative reported by us⁵⁹ and in an oxalate-bridged heterobimetallic chain.⁶⁶ A comparison of structural parameters for **1** and $\{(\text{Me}_2\text{NH}_2)_2[\text{Fe}_2(\text{ox})_2\text{Cl}_4]\cdot\text{H}_2\text{O}\}_n$ (**1a**)⁵⁹ is reported in Table S1 in the Supporting Information. The methylammonium cations in **1** are linked to the chains by means of hydrogen

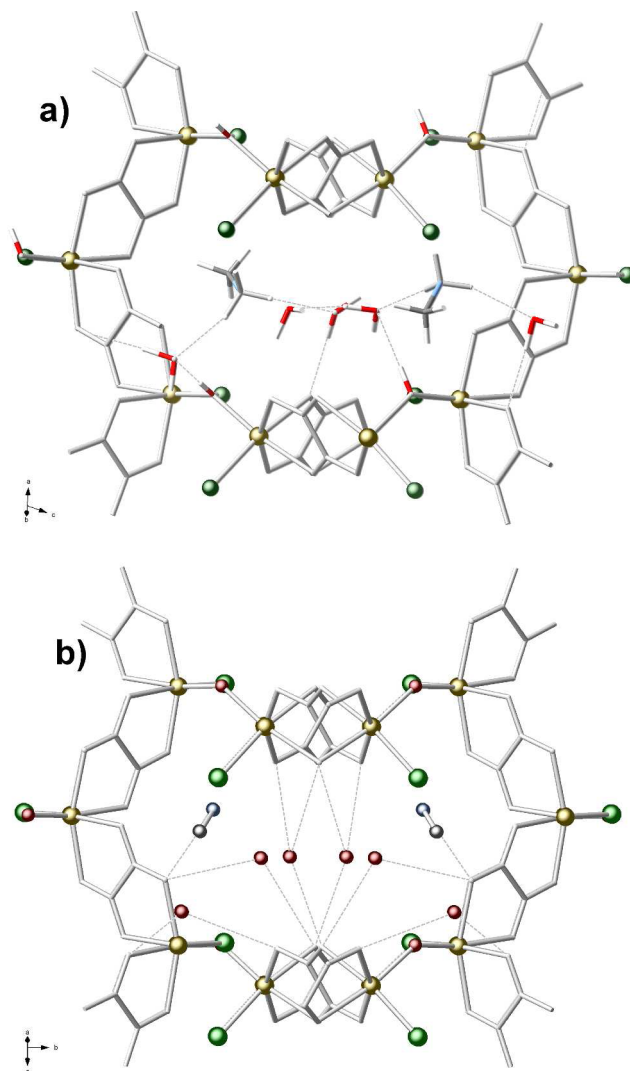


Figure 5. Crystal structures of 10-gon rings of the 3D anionic networks in **3** (a) and **4** (b) showing the slightly different positions of the cations and the water molecules located in the helical tunnels (dimensions of ca. $15.7 \times 8.7 \text{ \AA}^2$). The hydrogen atoms in **4** were not located and thus are not represented. Color code: iron (yellow), chlorine (green), oxygen of water molecules and hydroxo or oxo groups (red), nitrogen (pale blue), oxygen of the ligand and carbon (gray), hydrogen (pale gray).

bonds (see Figure 1), by direct means ($\text{N}(1)\cdots\text{O}(2)$), and also through crystallization water molecules that act as bridges ($\text{N}(1)\cdots\text{O}(3w)\cdots\text{O}(6)$ and $\text{N}(2)\cdots\text{O}(2w)\cdots\text{O}(1)$). Adjacent chains are linked by means of hydrogen bonds involving chloride ligands and water molecules ($\text{O}(1w)\cdots\text{Cl}(3)$) and the methylammonium cations ($\text{N}(1)\cdots\text{Cl}(1)$ and $\text{N}(2)\cdots\text{Cl}(3)$) (see Figures S2 and S3 in the Supporting Information).

Compound **1** is hygroscopic, and its crystals rapidly dissolve once separated from the mother liquor, affording microcrystals of **3** after 3 days (see the Experimental Section). A similar phenomenon has been observed even in the dimethylammonium derivative of formula $\{(\text{Me}_2\text{NH}_2)_2[\text{Fe}_2(\text{ox})_2\text{Cl}_4]\cdot\text{H}_2\text{O}\}_n$ ⁵⁹ but the process occurs much more slowly in this latter case. Thus, it is noticeable that both the MeNH_3^+ and the Me_2NH_2^+ cations allow the formation of a 1D compound with the same unique topology which can further react to transform into the oxo-bridged 3D species, hereunder described as compound **4**.

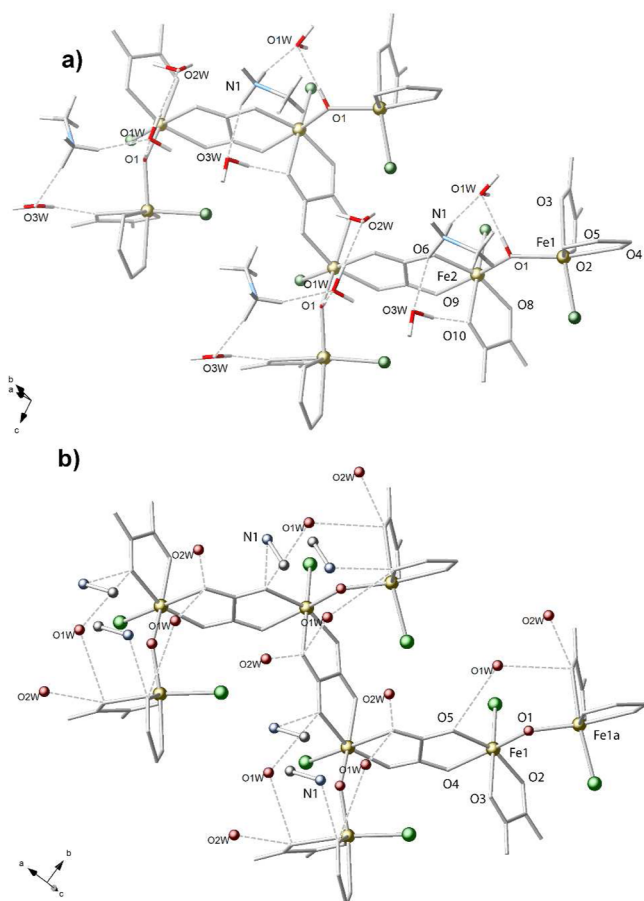


Figure 6. Local arrangement of the MeNH_3^+ ions and the water molecules in **3** (a) and **4** (b) where the hydroxo group is transformed into an oxo group and a water molecule is protonated. A disordered rearrangement of the cations and the protonated water molecules occurs in **4** most likely due to repulsive interactions between the H_3O^+ and MeNH_3^+ cations. Symmetry code: (a) $1 - x, -y, z$. Color code: iron (yellow), chlorine (green), oxygen of water molecules and hydroxo or oxo groups (red), nitrogen (pale blue), oxygen of the ligand and carbon (gray), hydrogen (pale gray).

In contrast, the common zigzag chains obtained with the Me_3NH^+ and Me_4N^+ counterions are stable and do not proceed toward the formation of more complex structures.

$K(\text{MeNH}_3)[\text{FeCl}_3(\text{ox})(\text{H}_2\text{O})]$ (**2**). This compound crystallizes in the $P2_1/m$ space group of the monoclinic system (see Table

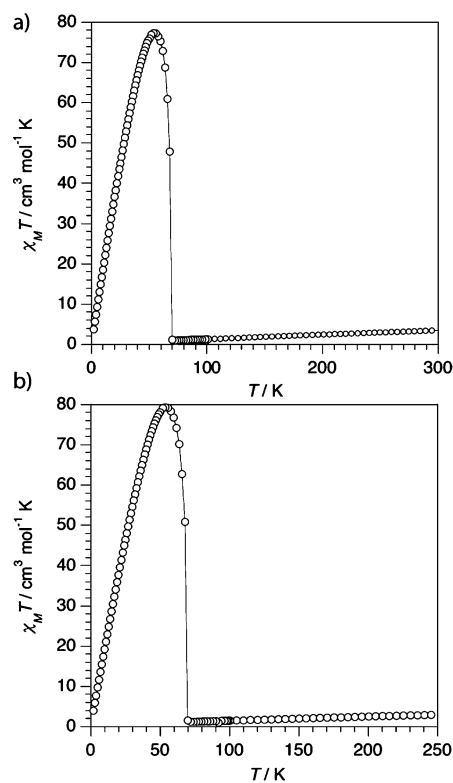


Figure 7. Temperature dependence of $\chi_M T$ for **3** (a) and **4** (b) under an applied dc field of 100 Oe.

1). Its structure is made up of mononuclear $[\text{FeCl}_3(\text{ox})(\text{H}_2\text{O})]^{2-}$ complex anions together with MeNH_3^+ and K^+ cations, which are held together by electrostatic forces, hydrogen bonds, and weak $\text{C}\cdots\text{H}\cdots\text{Cl}$ type interactions (Figure 2). The iron(III) ion of the $[\text{FeCl}_3(\text{ox})(\text{H}_2\text{O})]^{2-}$ unit is six-coordinate with a bidentate oxalate ligand, three chloride ions, and a water molecule building a distorted-octahedral surrounding. The asymmetric unit contains half of the mononuclear complex with the iron atom lying on a mirror plane. The values of the $\text{Fe}-\text{O}(\text{ox})$ (2.052(1) Å) and $\text{Fe}-\text{Cl}$ bond lengths (2.317(1)–2.333(6) Å) are in agreement with those reported for other similar compounds (Table S2 in the Supporting Information).^{28,67–70} The best equatorial plane around the Fe(1) atom is defined by the O(1)O(1a)Cl(2)Cl-(2a) set of atoms with the metal atom being 0.142(2) Å out of this plane.

Table 2. Selected Structural Data for **3** and **4** and Related Compounds

X	Fe–OH/Å	Fe–OH–Fe/deg	Fe $\cdots$ Fe ^a /Å	ref
MeNH_3^+ (3)	1.926(4)	132.3(2)	3.529(1)/	this work
	1.932(5)		5.429(1)	
EtNH_3^+	1.984(3)	132.4(2)	3.539(1)/	55
	1.882(3)		5.525(1)	
PrNH_3^+	1.934(7)	133.7(4)	3.550(2)/	56
	1.926(6)		5.457(2)	
X	Fe–O _{oxo} /Å	Fe–O _{oxo} –Fe/deg	Fe $\cdots$ Fe ^b /Å	ref
$\text{H}_3\text{O}^+/\text{MeNH}_3^+$ (4)	1.901(2)	133.9(4)	3.498(1)/	this work
			5.454(1)	
$\text{H}_3\text{O}^+/\text{EtNH}_3^+$	1.825(3)	136.0(5)	3.384(1)/	55
			5.482(1)	

^aIron–iron separation through the hydroxo/oxalate bridges. ^bIron–iron separation across the oxo/oxalate bridges.

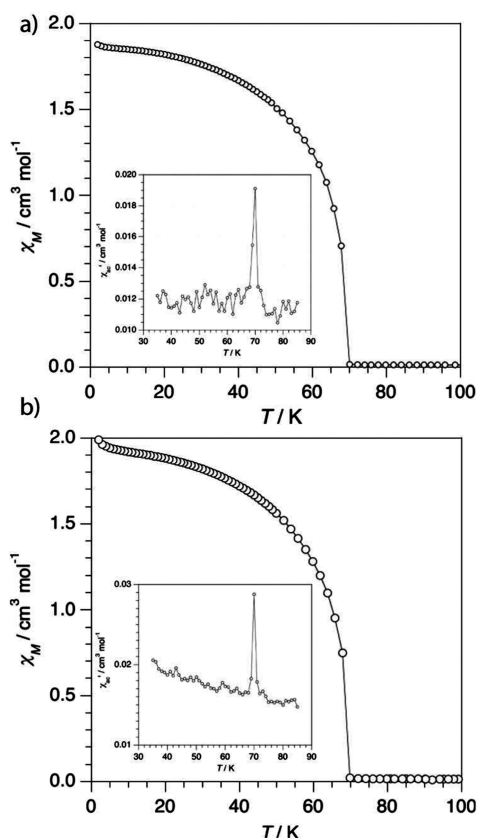


Figure 8. Field-cooled magnetization (FCM) for 3 (a) and 4 (b) (measured upon cooling within a field of 50 Oe). The insets show the temperature dependence of χ_M'' at 1000 Hz for 3 (a) and 4 (b). The solid lines are only guides for the eye.

The main distortion from the ideal octahedral geometry of the metal environment is due to the reduced bite angle of the oxalate ligand ($\text{O}(1)\text{--Fe}(1)\text{--O}(1a) = 79.03(7)^\circ$; symmetry code (a) $x, -y + 3/2, z$). The shortest intermolecular iron...iron separation is $5.471(2) \text{ \AA}$ ($\text{Fe}(1)\cdots\text{Fe}(1i)$; symmetry code (i) $2 - x, 1 - y, 1 - z$).

The potassium(I) ions in 2 are grasped by the complex anions through electrostatic interactions. Each univalent cation can be viewed as 10-coordinate with four peripheral oxalate oxygens, five chlorine atoms, and a water molecule describing a highly distorted bicapped square antiprism (Figure S4 in the Supporting Information). The values of the $\text{K}\cdots\text{Cl}$ and $\text{K}\cdots\text{O}$ interactions vary in the ranges $2.694(1)\text{--}3.011(2)$ and $3.352(1)\text{--}3.567(2) \text{ \AA}$ (Table S2 in the Supporting Information), respectively (Figure 2). The univalent alkaline cations are connected to each other through a single oxalato oxygen atom and/or double chloro bridges (Figure 3 and Figure S5 in the Supporting Information). The arrangement of the mononuclear iron(III) units and the K^+ ions leads to a 3D supramolecular motif (Figures 3 and 4), exhibiting channels which develop along the crystallographic c axis. The methylammonium cations are hosted in these channels, being fixed to the walls through $\text{N}\cdots\text{O}_{\text{ox}}$ and $\text{N}\cdots\text{H}\cdots\text{Cl}$ hydrogen bonds ($\text{N}(1)\cdots\text{O}(2) = 3.192(1) \text{ \AA}$ and $\text{C}\cdots\text{Cl} = 3.303(2) \text{ \AA}$) (Figure 4 and Figure S5).

$\{\text{MeNH}_3[\text{Fe}_2(\text{OH})(\text{ox})_2\text{Cl}_2]\cdot 3\text{H}_2\text{O}\}_n$ (3) and $\{(\text{H}_3\text{O})(\text{MeNH}_3)[\text{Fe}_2\text{O}(\text{ox})_2\text{Cl}_2]\cdot 2\text{H}_2\text{O}\}_n$ (4). Compounds 3 and 4 show a 3D MOF basically analogous to that found in previous compounds^{56–59} with the cations located in the anionic cages, the presence of an hydroxo bridge in 3 instead of the

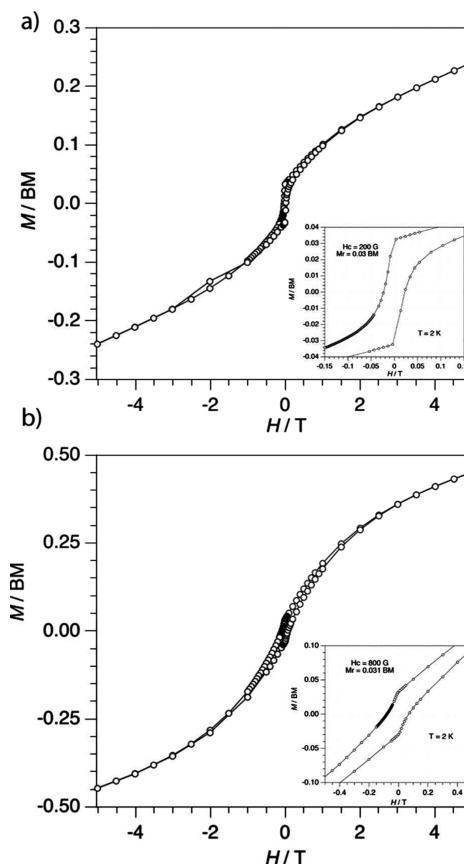


Figure 9. Field dependence for 3 (a) and 4 (b) of M at 2.0 K. The insets show the hysteresis loops in detail.

oxo bridge in 4 being the only difference. This structural feature merely causes slight modifications in the 3D framework (Figure 5a (3), Figure 5b (4), and Figure S6 in the Supporting Information (3)).

Compound 3 retains its framework and single crystallinity upon a proton-transfer reaction that occurs within the net, leading to the new species 4, where the hydroxo group is transformed into an oxo group and a water molecule of crystallization is protonated. The proton transfer is also followed by an almost disordered rearrangement of the cations and the protonated water molecules probably induced by repulsive interactions between the H_3O^+ and MeNH_3^+ cations (Figure 6). The general postulated mechanism for this single-crystal to single-crystal transformation is similar to that already reported in the case of the ethylammonium-containing compound.⁵⁸ In the present case with methylammonium as the cation, this process takes place quite rapidly, most likely due to the greater flexibility of the net in rearranging a smaller cation.

Each iron(III) ion in 3 and 4 is in a distorted-octahedral environment, being bonded to four oxygen atoms from two *cis* oxalate groups, a chlorine atom, and one oxygen atom of either a hydroxo (3) or an oxo group (4). The larger structural modifications when 3 is compared with 4 concern the values of the angles at the single hydroxo and oxo bridges, which are more bent in the former case ($132.3(2)^\circ$ (3) versus $133.9(4)^\circ$ (4)), and the $\text{Fe}\text{--O}(\text{bridge})$ bond length, which is shorter in the latter case ($1.926(4)$ and $1.932(5) \text{ \AA}$ (3) and $1.901(2) \text{ \AA}$ (4)). The values of the $\text{Fe}\text{--O}(\text{hydroxo})$ and $\text{Fe}\text{--O}(\text{oxo})$ bond lengths agree with those reported in the literature for parent 3D

iron(III) compounds (see Table 2).^{58–60} In addition, the Fe–O(ox) and the Fe–Cl bond distances (values varying in the range 1.926(4)–2.125(4) (3)/2.042(3)–2.135(3) Å (4) and 2.250(2)–2.282 Å in 3 versus 2.279(2) in 4, respectively) agree with those reported in the literature.^{58–60}

The reduced value of the bite angle of the oxalato ligand in 3 and 4 (minimum values of 79.4(2) and 79.07(9)°, respectively) is the main source of distortion of the iron(III) environment from the ideal octahedral geometry. The values of the iron–iron separation through the single hydroxo (3) or oxo (4) bridges as well as those through the bridging oxalato groups are given in Table 2.

Structural Overview: Role of the Cation. In light of these results, some suggestions may be made with regard to the formation of these types of compounds and the role of the cationic guest in the self-assembly process. The formation of the organized structure of 4 clearly involves (i) the self-assembly of simple building units such as 2 and (ii) a spontaneous irreversible proton-transfer process during the final stage that leads from 3 to 4. A more complicated task is to demonstrate whether the formation of the 3D open framework involves a building-up process from the 1D structure. The lack of evident structural correlations between the 1D starting material and the final product implies a different mechanism, wherein the 1D compound is most likely a secondary unstable product rather than a reaction intermediate.

On the other hand, the formation of 1 (a compound with a lifetime at room temperature of ca. 2 min when it is removed from the mother liquor) is new evidence of the chain-mediated self-assembly process leading to the 3D species. The temperature was found to be a key parameter with respect to the transformation of the intermediate. The chain complex dissolves slowly at 2 °C, and it transforms into the 3D hydroxo species 3 through a self-assembly process. At these low temperatures, an intramolecular proton transfer takes place slowly in 3, leading to the formation of the 3D oxo species 4. In turn, this solid to solid reaction rapidly occurs at room temperature.

The involvement of the chain complex in the self-assembly process could offer a possible explanation for the selectivity of the network toward the guest inclusion of the alkylammonium cations depending on their ability to form hydrogen bonds, unlike the alkaline cations. It could be suggested that the weak hydrogen bonds involving the protonated alkylamine molecules are more effective than the stronger ionic interactions provided by the alkaline cations toward the formation of such organized structures, the former being preferred in the self-assembling process when it is possible. Moreover, the isolation of the two unstable forms 1 and 3 has been possible only by using alkylammonium cations, suggesting their ability to temporarily “freeze” secondary products. It is well-known in the organic and biological fields that the occurrence of hydrogen bonds contributes to the stabilization of many reaction intermediates.^{71–74} The intermolecular hydrogen bonds where the alkylammonium cations and the crystallized water molecules are involved ensure the crystal cohesion of the chain compound and stabilize the hydroxo bridge in the 3D framework as well. Moreover, the process of the proton transfer from the hydroxo bridge to a crystallization water molecule in the solid phase is assisted by the hydrogen-bonding interactions with the cations, as observed in 2 and in the ethylammonium derivative. Such a support cannot be present in the case where the alkaline cations are the guests, most likely making this species more unstable

and speeding up the transformation. This is why the isolation of a hydroxo-bridged compound with an alkaline cation as guest is prevented.

Magnetic Properties of 3 and 4. Overall, the magnetic properties of 3 and 4 are typical of spin-canted compounds (Figures 7–9). The thermal variation of the $\chi_M T$ products for 3 and 4 (χ_M being the magnetic susceptibility per two iron(III) ions) under an applied dc field of 100 Oe is shown in Figure 7. At room temperature, the values of $\chi_M T$ are equal to 3.92 and 4.25 cm³ mol^{−1} K for 3 and 4, respectively. They are much smaller than those expected for two magnetically isolated high-spin Fe(III) ions (8.75 cm³ mol^{−1} K for $S = 5/2$ with $g = 2.0$), a feature which is as expected because of the strong antiferromagnetic interactions previously observed between iron(III) ions across oxo, hydroxo, and oxalato bridges.^{56–59,66,75–88} When the samples are cooled, $\chi_M T$ decreases and attains minima at 72 (3) and 73 K (4). Below these temperatures, $\chi_M T$ increases abruptly to reach maxima at 57 (3) and 53 K (4) ($\chi_M T = 78.20$ and 80.2 cm³ mol^{−1} K, respectively), and it further decreases linearly with T in the very low temperature range, tending to vanish in both cases.

Interestingly, 3 and 4 behave as magnets through spin canting, showing a high critical temperature (T_c) of 70 K in both cases. These magnetic orderings are confirmed by the combination of both the measurement of the temperature dependence of the field-cooled magnetization (FCM) curves, which show a very fast increase below 70 K, on the one hand (Figure 8) and the sharp frequency-independent maxima at 70 K in the χ''_M vs T plots (inset of Figure 8) on the other hand. Such a high T_c value is the same as that observed one in related oxalate-/oxo- and oxalate-/hydroxo-bridged iron(III) 3D compounds of formulas $\{(H_3O)(EtNH_3)[Fe_2O(ox)_2Cl_2] \cdot H_2O\}_n$,⁵⁸ $\{EtNH_3[Fe_2(OH)(ox)_2Cl_2] \cdot 2H_2O\}_n$,⁵⁸ and $\{PrNH_3[Fe_2(OH)(ox)_2Cl_2] \cdot 2H_2O\}_n$.⁵⁹

The magnetization versus H plots for 3 and 4 at 2.0 K exhibit linear increases to reach maximum values of ca. 0.25 (3) and 0.42 μ_B (4) at 5 T (Figure 9). The hysteresis loops of 3 and 4 at 2.0 K show values of the coercive field (H_c) and remnant magnetization (M_r) of 200 (3) and 800 Oe (4) and 0.030 (3) and 0.031 μ_B (4), respectively (inset of Figure 9). From the saturation value of the magnetization of the canting (ca. $M_c = 0.017$ (3) and 0.018 μ_B (4)) (Figure 9) and the expected value for two spin $S = 5/2$ ($M_S = 10 \mu_B$),⁸⁹ values of the canting angle (α) of ca. 0.097 (3) and 0.103° (4) are calculated.⁹⁰ The spin-canting phenomenon in the low-temperature domain for 3 and 4 has its origin in the antisymmetric exchange, which is compatible with the absence of inversion center in their crystal structures (the isotropic character of the high-spin iron(III) ruling out the magnetic anisotropy as a source of spin canting).^{91–93}

CONCLUSIONS

The synthesis and crystal structures of four new oxalate-based iron(III) complexes have been reported. Furthermore, the magnetic properties of the 3D compounds 3 and 4 have been investigated, both of them being spin-canted systems and exhibiting magnetic ordering with a value of T_c of 70 K. The isolation of these unprecedented methylammonium-containing compounds (1–4) has been possible exclusively in the presence of alkaline cations in the solution of the reaction mixtures, suggesting a probable templating effect and giving new insights into the possibility of obtaining less favored products or isolating very unstable species (as 1) by further

changing the reaction conditions. In particular, it can be suggested that the presence of electrolytes affects the chemical equilibria, leading to the formation of minor products. Thus, these compounds exhibit a high selectivity toward the alkylammonium cation as guest, the alkaline cations being excluded from the network. This preference could be most likely caused by a host–guest recognition process that involves the formation of multiple hydrogen-bonding interactions.

■ ASSOCIATED CONTENT

■ Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: [10.1021/acs.inorgchem.6b01769](https://doi.org/10.1021/acs.inorgchem.6b01769).

Additional tables and figures as described in the text (PDF)

X-ray data for 1 (CCDC 1495414) (CIF)

X-ray data for 2 (CCDC 1495415) (CIF)

X-ray data for 3 (CCDC 1495416) (CIF)

X-ray data for 4 (CCDC 1495417) (CIF)

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Notes

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